TCAD-Based Assessment of the Lateral GAA Nanosheet Transistor for Future CMOS

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***Abstract*— A physical assessment of the lateral gate-all-around (GAA) nanosheet transistor (NSFET) at the G40M16 node (gate length = 12 nm projected for 2028) of the newly defined beyond-Moore International Roadmap for Devices and Systems [~5 nm node of the predecessor International Technology Roadmap for Semiconductors (ITRS)], supported by 3-D numerical device simulations and based on a proposed per-footprint scheme, is presented. The traditional evaluation scheme to gauge the performance of the transistor "per effective channel width" is shown to be improper due to pervasive bulk inversion. Bulk inversion, along the width of the nanosheet and at its ends, obfuscates the dependence of current on sheet/channel width and undermines the presumed added benefit of short-channel effect (SCE) control in the GAA device. The NSFET provides a flexible sheet width, as opposed to that of the FinFET with discrete fins, but the allowed range of sheet width is limited due to significant parasitic capacitance at a relatively narrow width and degraded SCE control and increased resistance for desired wide width. Furthermore, the nanosheet (NS) device density is undermined at a narrow width, especially for decreasing pitch. The FinFET is shown to be a favorable alternative to the NSFET even at the G40M16 node.**

***Index Terms*— Bulk inversion, device density, effective channel width, FinFET, footprint, gate capacitance, NSFET, S-D series resistance, short-channel effects (SCEs).**

I. Introduction

RECENT articles and documents [1]–[5] have promoted the horizontal lateral gate-all-around (GAA) nanosheet transistor (NSFET) as a viable candidate to replace the FinFET at and beyond the 5-nm CMOS node of the International Technology Roadmap for Semiconductors (ITRS). Such promotion is based mainly on potential superior short-channel effects (SCEs) and current drive in the NSFET relative to the FinFET, with an adjustable device width that could be optimized for different applications.

The potential SCE superiority of the NSFET is due to the GAA structure [6] shown in Fig. 1, which is beneficial if the NSFET thickness (t_{Si}) is ultrathin like that of a FinFET and the NSFET width (w_{Si}) is sufficiently narrow to effect four charge-coupled gates, as opposed to just two for the FinFET [7]. If w_{Si} is too wide, then the end gates are not significant, due in part to bulk inversion [7]–[9], and the NSFET is virtually a horizontal double-gate (DG) FinFET with worse parasitics. So, the lateral, or vertical, GAA structure, which is expected to be the mainstream device through 2034 [10], is basically a (DG) FinFET for the normally wide w_{Si}. The bulk inversion limits the gate controllability in the undoped FinFETs, even in the on-condition [8]. The SCE superiority, for narrow-w_{Si} NSFETs, is presumed to be manifested by a higher ON-current–OFF-current ratio ($I_{\mathrm{ON}}/I_{\mathrm{OFF}}$) that would follow from a reduced subthreshold swing (SS) and drain-induced barrier lowering (DIBL). However, other mechanisms, like the bulk inversion and the general invalidity of the effective channel-width assumption that is commonly made, $W_{\mathrm{eff}} = 2 \times (w_{\mathrm{Si}} + t_{\mathrm{Si}})$ [1], tend to limit the actual benefit realized. When considering the impact from quantum mechanical (QM) carrier confinement, the common assumption of W_{eff} becomes even more ambiguous [11]. We give herein a physical assessment of the presumed NSFET advantages over the FinFET at the G40M16 node (gate length $L_{\mathrm{g}} = 12$ nm projected for 2028 at gate pitch = 40 nm and metal-0 pitch = 16 nm) of the newly defined beyond-Moore International Roadmap for Devices and Systems [10] based on 3-D numerical device simulations, including those related to speed and density, as well as electrostatics.

II. Device Simulations

We use Sentaurus TCAD [12] for the 3-D simulations. We neglect the QM effects [7], as t_{Si} can be relaxed due to good SCE control to avoid significant threshold voltage (V_{t}) variations due to QM effects in very thin t_{Si}. When considering moderate or wide w_{Si}, the QM effects are comparable for both the NSFET and FinFET. For the latter to have suppressed SCEs without significant QM effects, we need $t_{\mathrm{Si}} \sim$ effective channel length $(L_{\mathrm{eff}})/2 > \sim 4$ nm [7]. When w_{Si} is narrow, the QM effects of the NSFET approach those of the GAA nanowire [13], for which an approximate 2-D extension of the 1-D QM analysis in [14] shows that t_{Si} must be at least $\sim 1.3\times$ thicker (1.3 nm × 4 nm) to avoid excessive V_{t}

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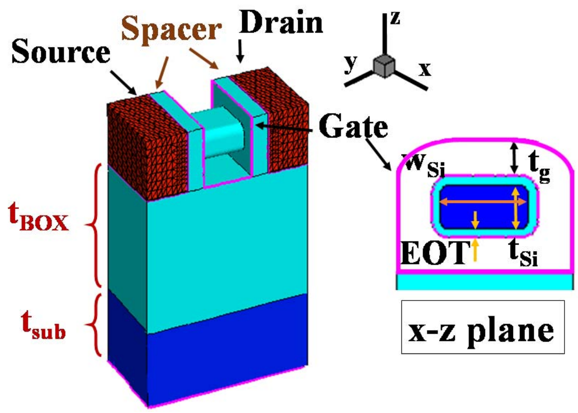

Fig. 1. 3-D view and channel cross section of a horizontal single-sheet NSFET.

variations. In fact, supplemental Sentaurus simulations of the two devices with QM effects validate this simplification. These criteria validate this QM simplification for the NSFET with $t_{\mathrm{Si}} = 5$ nm.

Fermi–Dirac statistics and classical drift-diffusion transport using the Philips unified mobility model, with surface scattering and high-field velocity saturation, are selected for the simulations; velocity overshoot is accounted for by assuming an effective saturated drift velocity $\upsilon_{\mathrm{sat(eff)}} = 3 \times 10^7$ cm/s [15] greater than υ_{sat}. The source and drain (S/D) regions are assumed *in situ* doped at 1×10^{20} cm^{-3}. The target contact resistivity is 4×10^{-9} Ω-cm^2 based on reported resistivity values of 5×10^{-9} Ω-cm^2 [16], and even as low as 1×10^{-9} Ω-cm^2 with integrated Ti, TiN, and Co fabrication [17]–[19]. Besides contact resistivity, the actual contact resistance is also dependent on the S/D structure and area subject to epitaxial process and contact pitch doable at the 5-nm technology node [18], although the S/D structure of our work is assumed homogeneous in a rectangular solid and the contacts wrap around the outer four faces, as shown in Fig. 1 (shaded S/D surface) [20], [21]. Overall, our stated model assumptions constitute a good representation of the G40M16 lateral NSFET [10].

For initial assessment of the basic NSFET, we use an n-channel single-sheet horizontal device, as shown in Fig. 1, having variable body/channel dimensions, w_{Si} and t_{Si}. The equivalent oxide thickness (EOT), spacer width, and gate thickness (t_g) are fixed at 1, 4, and 4 nm, respectively. Table I provides a summary of the device parameters assumed throughout the article. The single-sheet structure is designed on the silicon-on-insulator (SOI) substrate, with undoped ultrathin body (UTB)/channel, metal gate on classical SiON, gate–source/drain (G-S/D) underlap, and no lattice strain [13]. Such a pragmatic design effects a good tradeoff between CMOS performance and process complexity for nanoscale gate lengths [7]. We assume $L_g = 12$ nm [1] for the G40M16 node [10].

Later, for a performance comparison with the FinFET, we expand the domain to include three stacked nanosheets, and define the FinFET domain to include a number of vertical fins that fit within the footprint width of the NSFET stack.

TABLE I
PARAMETERS OF THE SIMULATED NSFET AND FINFET COUNTERPART

	NSFET	FinFET
Gate length (L_g)	12nm	
Effective channel length (L_{eff})	16nm	
UTB/sheet thickness (t_{Si})	5nm	
BOX thickness (t_{BOX})	100nm	
Equivalent oxide thickness (EOT)	1nm	
Length of spacer (LSP)	4nm	
Gate pitch	30nm	
Gate thickness (t_g)	4nm (top/side/bottom) 8nm (between sheets)	4nm (top/side) 8nm (between fins)
Sheet width/fin height (w_{Si}/h_{Si})	50nm	40nm (same as NS stack)
Stacked sheets (n_f) /number of parallel fins	3	4
Footprint width	60nm	
Supply voltage (V_{DD})	0.6V	
Gate work function (WF)	4.38eV	

III. SINGLE-SHEET DEVICE RESULTS

Fig. 2 shows the predicted SCEs (DIBL and SS) of the single-sheet NSFET ($L_{\mathrm{eff}} = 16$ nm defined by the G-S/D underlap) at V_{GS} and $V_{\mathrm{DS}} = 0.6$ V for w_{Si} increasing from 5 nm (which defines a nanowire with $t_{\mathrm{Si}} = 5$ nm [13]) to 65 nm (which defines a virtual DG FinFET). Note the SCE superiority of the NSFET for narrow w_{Si} over the FinFET, but loss of that advantage for increasing w_{Si}; for $w_{\mathrm{Si}} > \sim 4 \times t_{\mathrm{Si}} = 20$ nm, the advantage is virtually gone. This implies an upper limit for w_{Si} if the NSFET SCE superiority is to be exploited. The inset of Fig. 2 shows the electron density profiles of the NSFET at $w_{\mathrm{Si}} = 50$ nm and $t_{\mathrm{Si}} = 5$ nm (along the xz plane in Fig. 1) for $V_{\mathrm{GS}} = 0.6$ V and $V_{\mathrm{DS}} = 0$ V, where the bulk inversion is significant in the entire silicon channel [20]. To better quantify this w_{Si} limit and its dependence on t_{Si}, predicted w_{Si} versus t_{Si} design space for assumed small SCEs (DIBL = 40 mV/V and SS = 70 mV/dec), a la that for FinFETs [7], is shown in Fig. 3. Note for decreasing w_{Si} how the needed NSFET t_{Si} becomes thicker than that of the FinFET, but not until $w_{\mathrm{Si}} \sim L_{\mathrm{eff}} = 16$ nm. Here the efficacy of the two end gates of the NSFET is limited some by bulk inversion supported by the top and bottom gates on the sheet/channel [7]–[9]. Samsung suggests that its multibridge-channel FET (MBCFET) [22], which basically is an NSFET, is the "most feasible CMOS solution beyond the FinFET" because of near-ideal SS, higher current with a larger W_{eff} for a reference footprint, and design flexibility with variable nanosheet width. However, all these advantages require $w_{\mathrm{Si}} < \sim L_{\mathrm{eff}}$, which seems impractical.

Fig. 4 shows the predicted OFF-state currents versus w_{Si} at $V_{\mathrm{DD}} = 0.6$ V. The I_{OFF} plots reflect [beyond the $\log(I_{\mathrm{OFF}})$ curvature] the SCE influence of narrow w_{Si}; I_{OFF} actually

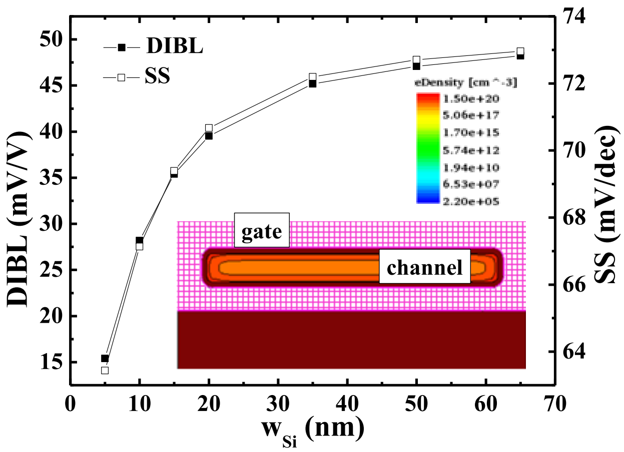

Fig. 2. Predicted DIBL and SS of NSFET versus w_{Si} at V_{GS} and $V_{\mathrm{DS}} = 0.6$ V, $L_{\mathrm{eff}} = 16$ nm, $t_{\mathrm{Si}} = 5$ nm, and WF $= 4.38$ eV. The inset is the electron density profile of the NSFET at $w_{\mathrm{Si}} = 50$ nm and $t_{\mathrm{Si}} = 5$ nm (along the *xz* plane shown in Fig. 1) for $V_{\mathrm{GS}} = 0.6$ V and $V_{\mathrm{DS}} = 0$ V.

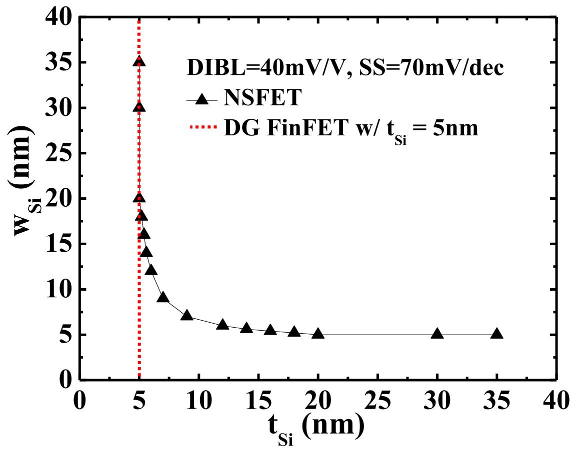

Fig. 3. Predicted NSFET design space (w_{Si} versus t_{Si}) for the noted SCEs compared to that of the FinFET counterpart at $L_{\mathrm{eff}} = 16$ nm, WF $=$ 4.38 eV, and $V_{\mathrm{GS}} = V_{\mathrm{DS}} = 0.6$ V.

decreases quite dramatically for decreasing w_{Si}, reflecting improved SCEs due to the NSFET-end gates [13]. Included in Fig. 4 is the I_{OFF} versus w_{Si} plot for a longer $L_g = 100$ nm device to clearly illustrate, by contrast, this SCE improvement. However, we stress that this benefit is significant only for a narrow w_{Si} ($\sim L_{\mathrm{eff}} < \sim 4 \times t_{\mathrm{Si}}$), consistent with the discussions shown in Figs. 2 and 3.

The predicted ON-state currents versus w_{Si} at $V_{\mathrm{DD}} = 0.6$ V in Fig. 5 show virtual linearity for all w_{Si} (since the SCEs are well controlled here for thin enough t_{Si}); they show $I_{\mathrm{ON}} \propto a \times w_{\mathrm{Si}} + b$, where "$b$" (the I_{on}-axis intercept) can be inferred as a current component governed by the NSFET-edge gates. The commonly assumed $W_{\mathrm{eff}} = 2(w_{\mathrm{Si}} + t_{\mathrm{Si}})$ is clearly not valid (due to bulk inversion), analogous to that for FinFETs [7]. Note that this effect of bulk inversion relates to the fact that the DG-FinFET current is less than twice the current of the single-gate counterpart [7], i.e., the slope "a" in I_{ON} expression above is not 2. Fig. 5 shows an ideal I_{ON} versus w_{Si} curve, without S/D series resistance (R_{SD}) (i.e., with the contact resistivity set to zero), to reveal R_{SD} effect. It stems mainly from the S/D contact discussed in Section II, and is a concern for wide-w_{Si} stacked-nanosheet (NS) devices

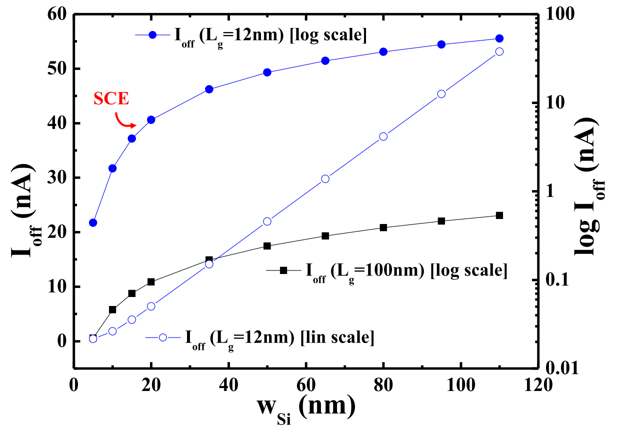

Fig. 4. Predicted I_{OFF} versus w_{Si} for the NSFET with short and long gate lengths, reflecting the SCE influence of narrow w_{Si} at $V_{\mathrm{DD}} = 0.6$ V and WF $= 4.38$ eV.

in which the contact does not wraparound each sheet [23], as opposed to the ideal wraparound contact in our single-sheet device here. Such increased resistance can also be observed in a multiple-fin FinFET with tall fins since the contact does not wraparound each fin due to the extremely scaled fin pitch. Note also that I_{ON} decreases further for decreasing w_{Si} due to the SCE effect (i.e., increased V_t), even though SS is improved. To get higher I_{ON} from the SS benefit, the gate work function (WF) would have to be lowered to reduce V_t.

To provide more evidence regarding W_{eff}, the four-gate NSFET device is converted to a two-gate device with no active end gates (which is virtually the DG FinFET), as shown in the inset of Fig. 6. The 4G versus 2G I_{ON} comparison versus w_{Si} shown in Fig. 6 is clearly inconsistent with the common W_{eff} assumption noted earlier. For example, for $w_{\mathrm{Si}} = 15$ nm, I_{ON} improvement from the end gates is about 13% (from Fig. 6), whereas the W_{eff}-based improvement is 33%. This discrepancy worsens for a narrower w_{Si}. For a wider w_{Si}, the end gates are virtually insignificant. With I_{ON} versus w_{Si} linearity shown in Fig. 5, W_{eff} model is very misleading with regard to the current-per-footprint metric.

IV. Stacked-NSFET Results With Comparison to FinFET

A three-sheet stacked NSFET is assumed for comparison with a multiple-fin FinFET, as shown in Fig. 7. Note that since the NSFET in this work is designed with SOI, the FinFET counterpart is too, which means that the S-D punchthrough leakage that plagues the bulk-Si FinFET is eliminated [7]. For both devices, $L_g = 12$ nm ($L_{\mathrm{eff}} = 16$ nm) and $t_{\mathrm{Si}} = 5$ nm. The sheet width and number of parallel fins (n_f) are varied for performance comparison, with the footprints of both devices kept equal by setting n_f based on w_{Si}. For high-performance design, the gate WF of the NSFET is set to 4.38 eV to get $I_{\mathrm{OFF}} =$ 40 nA for $w_{\mathrm{Si}} = 50$ nm at $V_{\mathrm{DD}} = 0.6$ V. The defined footprint will hold four parallel FinFETs with fin pitch of 15 nm [13], fin height $h_{\mathrm{Si}} = 40$ nm defined by the height of the stacked NSFET, and $I_{\mathrm{OFF}} = 10$ nA in each fin to get the total four-fin I_{OFF} of 40 nA as in the NSFET. To get $I_{\mathrm{OFF}} = 10$ nA for each fin, WF is tweaked up a bit (only 6 meV).

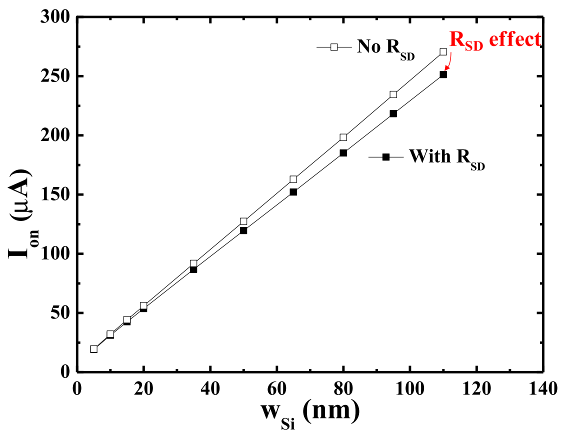

Fig. 5. Predicted I_{ON} versus w_{Si} for the L_g = 12 nm NSFET at V_{DD} = 0.6 V and WF = 4.38 eV. The noted R_{SD} effect is a concern for large w_{Si} in a stacked-NS device.

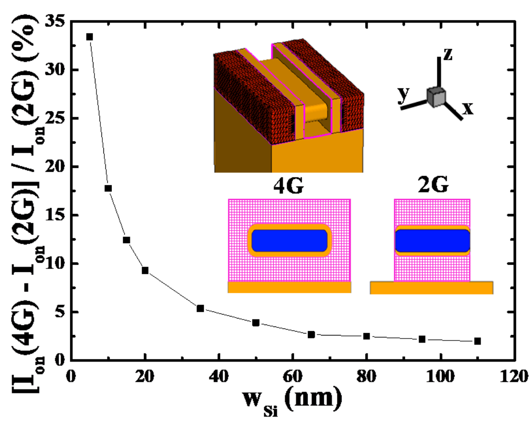

Fig. 6. 4G versus 2G I_{ON} comparison for the L_g = 12 nm NSFET at V_{DD} = 0.6 V and WF = 4.38 eV, showing the limited I_{ON} improvement from the end gates with decreasing w_{Si}.

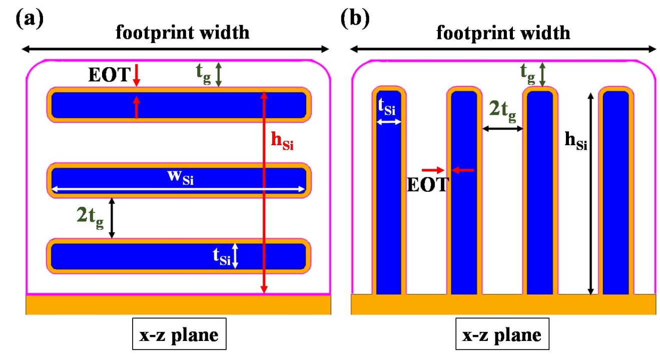

Fig. 7. 2-D *xz* cross sections of (a) stacked NSFET with sheet width w_{Si} and thickness t_{Si} and (b) multifin FinFET in the same footprint width with fin height h_{Si} and thickness t_{Si}.

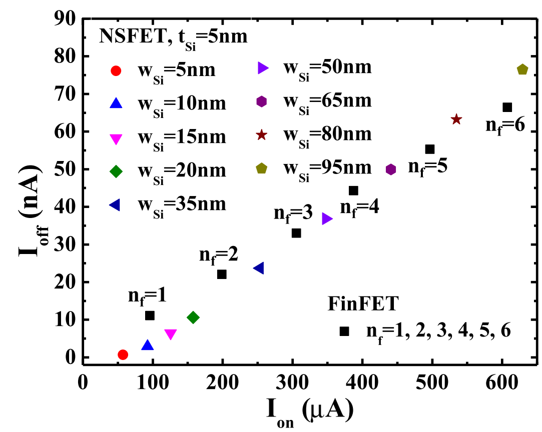

Fig. 8. Predicted I_{OFF} versus I_{ON} for the stacked NSFETs at L_g = 12 nm, t_{Si} = 5 nm, and w_{Si} = 5–95 nm, and multifin FinFET counterparts for n_f = 1–6 fins at V_{DD} = 0.6 V.

One would intuitively expect that the stacked NSFET I_{ON} is proportional to the stack number. In fact, our simulation results suggest reduced stacked-NSFET I_{ON}($< 3\times$ the single-sheet I_{ON}). Such I_{ON} discrepancy becomes even worse as w_{Si} increases due to the increased R_{SD} for wide w_{Si} discussed earlier. Higher parasitic resistance is inherent in the stacked structure since the metal contacts only wraparound the outer S/D perimeter, and the metal contact on top of the S/D region is more effective for the single-sheet device. If the contacts wrapped around the individual sheets, which is actually prohibitive in the NSFET processing, then R_{SD}-based reduction of I_{ON} could be eliminated. The associated NSFET performance loss is shown in Fig. 8 by predicted I_{OFF} versus I_{ON} for stacked NSFETs (w_{Si} = 5–65 nm) and multifin FinFET counterparts (n_f = 1–5 fins). At narrow w_{Si}, the NSFET is like a GAA nanowire FET that can deliver comparable I_{ON} with smaller I_{OFF} due to superior electrostatics. The advantage of NSFETs in I_{ON}/I_{OFF} clearly diminishes as w_{Si} increases.

The predicted ON-state gate capacitance (C_G, including parasitic G-S/D capacitance associated with the sheet ends and spacings) of the stacked NSFET is plotted, normalized by w_{Si}, as shown in Fig. 9(a), versus w_{Si}. Note how C_G/w_{Si} increases with decreasing w_{Si}. This nonlinearity clearly reflects a parasitic G-S/D component of C_G defined by the sheet-spacing geometry at the ends of the sheets that is independent of w_{Si}. The inset of Fig. 9(a) shows the stacked device and indicates how these components depend on t_{Si} for the fringing capacitance of sheet ends (green symbols), and on the sheet spacing for the inner-spacer induced capacitance (white symbols), but not on w_{Si}, implying that narrow w_{Si} is problematic. More complex processes such as reduced vertical-nanowire separation [24] and an inner-spacer technology [25] are needed to reduce parasitic capacitance. The FinFET avoids this issue via tall fins that render parasitic capacitance associated with the (top) gated fin edge insignificant.

The predicted intrinsic speed ($C_G V_{DD}/I_{ON}$) of the stacked NSFET, versus w_{Si}, is also plotted and shown in Fig. 9(a). Note that the speed improves with increasing w_{Si}, which implies that it is governed mainly by the noted w_{Si}-independent parasitic component of C_G becoming less significant as w_{Si} is widened. But, when w_{Si} is excessively widened, the S/D resistance (discussed in Section III) increases due to longer current paths from contact surfaces to each sheet (channel) of the stacked NSFET, as shown in the inset of Fig. 9(a). To provide further quantitative comparison, Fig. 9(b) shows a comparison of $C_G V_{DD}/I_{ON}$ for multifin FinFET and stacked NSFET at w_{Si} = 20–65 nm. The narrow NSFETs suffer from the parasitic capacitance effects. Nevertheless, the speed is governed mainly by C_G and not influenced significantly by

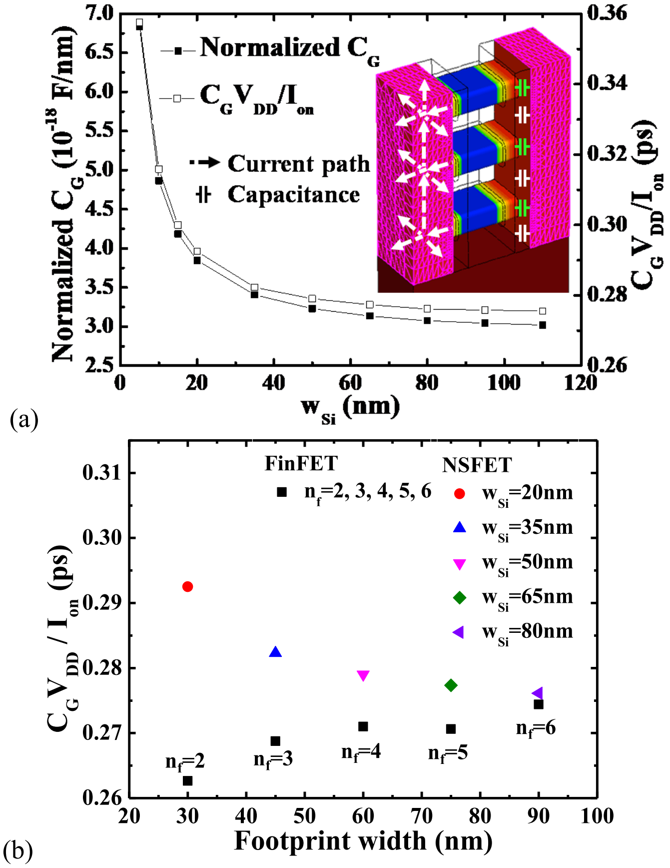

Fig. 9. (a) Normalized gate capacitance (per w_{Si}) and intrinsic delay time of the stacked NSFET versus w_{Si}. The inset shows the stacked-device structure, and indicates w_{Si}-independent parasitic capacitance components defined by geometries of the sheet ends (green symbols) and spacings, i.e., inner spacer (white symbols). It also shows S/D series-resistance paths (white arrows) that are longer for wider w_{Si}. (b) Predicted $C_G V_{DD}/I_{ON}$ for the multifin FinFET and the stacked NSFET at $w_{Si} = 20$–80 nm for $V_{DD} = 0.6$ V.

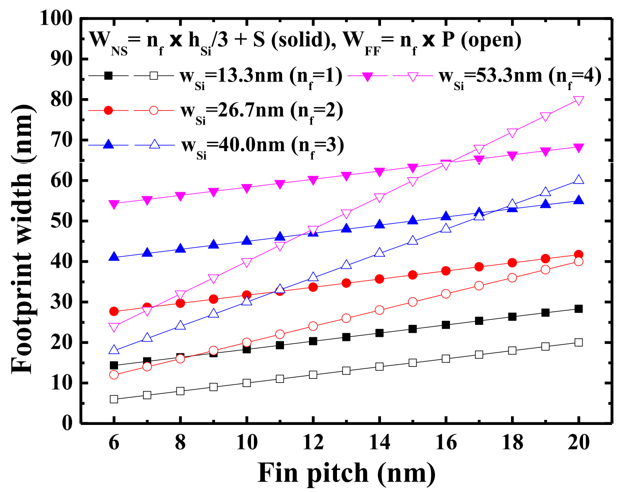

Fig. 10. Conceptual comparison of device densities (footprint width versus fin pitch) for NSFETs and FinFETs (widths W_{NS} and W_{FF}, respectively) at $V_{DD} = 0.6$ V and WF $= 4.38$ eV.

R_{SD}. The predicted trend here is that wide w_{Si} is optimal for speed, even though improved SCE control is sacrificed and any I_{ON} benefit from the nanosheet ends is negligible. For reasonably wide w_{Si} though, the NSFET is virtually a DG FinFET but with higher R_{SD} and C_G, and slower speed.

Finally, the FinFET appears to offer superior device density unless the NSFET width is moderately wide. For a specified drive current (defined by w_{Si} and n_f), the footprint widths (W_{NS} and W_{NF}) of the two devices shown in Fig. 7 can be estimated in terms of the fin pitch (P) and spacing ($S = P - t_{Si}$) [26]; S is the spacing between the stacked NSFETs and FinFETs. We find that the NSFET width ($W_{NS} = n_f \times h_{Si}/3 + S$) is larger than that of the FinFET width ($W_{FF} = n_f \times P$) unless w_{Si} (i.e., W_{NS}–S) is wider than that needed to exploit the SCE benefit, as shown in Fig. 10. w_{Si} required increases with decreasing P, thereby increasing R_{SD} and possibly reducing speed.

V. Conclusion

We have stressed four new insights regarding the lateral NSFET that undermine its lofty expectations for G40M16 CMOS and beyond.

1) The traditional use of effective channel width for the planar MOSFET is invalid for the NSFET (as it is for the FinFET [7]) due to prevalent bulk inversion. Although the adjustable sheet width overcomes the fin quantization of the FinFET, there are considerable limitations.
2) Also due to bulk inversion, the SCE benefit afforded by the end gates of the NSFET is negligible unless $w_{Si} < \sim L_{eff}$.
3) Excessive parasitic capacitance and resistance place, respectively, lower and upper limits on w_{Si} for NSFET speed competitiveness with the FinFET.
4) The NSFET footprint tends to be larger than that of the FinFET unless w_{Si} is moderately wide, too wide to exploit any SCE benefit. These insights, and the complex NSFET processing [1], [27], suggest that the NSFET is not preferable. The FinFET, with simpler processing, is comparable if not better in overall performance and device density, and is a favorable alternative.

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